

TROMSO, Norway

WMO: 010260

Lat: 69.6536N

Lon: 18.9367E

Elev: 115

StdP: 99.96

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-10.1	-16.6	0.9	-9.3	-14.8	1.0	-8.8	11.8	-1.0	10.8	-1.1	2.2	200	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	21.4	14.5	19.4	13.5	17.5	12.6	15.3	20.2	14.1	18.4	13.2	16.8	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	9.4	17.0	12.0	8.8	16.1	11.1	8.3	15.1	43.2	20.4	40.0	18.5	37.4	16.9	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.4	-13.3	24.9	2.1	2.8	-14.8	26.9	-16.1	28.6	-17.3	30.1	-18.8	32.1		
(5)			-14.0	17.0	2.0	1.7	-15.4	18.2	-16.6	19.2	-17.7	20.2	-19.2	21.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-3.5	-3.2	-1.9	1.6	6.0	9.0	12.6	11.6	8.1	3.3	-0.1	-1.5		
	DBStd	6.45	3.79	3.86	3.32	3.01	3.40	3.13	3.10	2.81	3.02	3.45	3.26	3.43		
	HDD10.0	2553	417	369	368	251	132	56	8	14	71	208	302	356		
	HDD18.3	5397	676	602	626	501	382	281	181	211	306	466	552	614		
	CDD10.0	201	0	0	0	0	9	25	88	63	14	1	0	0		
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0		
	CDH23.3	23	0	0	0	0	1	2	17	3	0	0	0	0		
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0		
(14) Wind	WSAvg	3.4	4.1	4.1	3.9	3.2	3.1	2.8	2.5	2.5	2.8	3.3	3.8	4.1		
(15) Precipitation	PrecAvg	973	94	88	64	54	43	50	66	72	105	130	102	104		
	PrecMax	1470	281	217	183	134	130	104	122	166	211	315	243	248		
	PrecMin	661	31	2	8	4	7	5	12	14	19	17	12	6		
	PrecStd	206	52	47	41	28	25	26	27	41	50	67	47	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	5.6	6.9	12.4	20.2	22.0	25.6	23.1	17.6	13.4	8.9	6.9		
		MCWB	3.2	3.0	2.8	6.8	11.6	12.9	17.1	15.4	12.2	9.6	6.2	4.2		
	2%	DB	4.4	4.6	5.2	9.2	16.4	18.8	22.4	20.1	15.7	11.0	6.8	5.5		
		MCWB	2.4	2.5	2.3	5.1	9.7	12.2	15.3	14.2	11.4	7.9	4.7	3.5		
	5%	DB	3.2	3.6	4.0	7.4	13.7	16.6	20.3	17.7	14.1	9.5	5.7	4.3		
		MCWB	1.5	1.8	1.8	4.0	8.2	11.2	14.3	13.3	10.4	7.2	3.6	2.3		
	10%	DB	1.9	2.4	3.0	5.9	11.5	14.5	18.4	16.1	12.6	8.2	4.6	3.3		
		MCWB	0.3	0.9	1.0	2.9	7.1	10.1	13.4	12.6	9.7	6.3	2.9	1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	4.0	3.6	7.2	12.2	14.3	17.7	16.5	13.1	9.9	7.0	5.1		
		MCDB	5.0	5.1	5.4	11.6	19.5	20.4	23.8	20.4	16.7	12.8	8.5	6.1		
	2%	WB	2.8	2.7	2.9	5.5	10.1	12.8	16.0	14.7	11.7	8.4	5.1	3.7		
		MCDB	4.1	4.1	4.7	8.6	15.8	17.8	21.6	18.7	14.6	10.6	6.4	5.4		
	5%	WB	1.6	2.0	2.0	4.3	8.6	11.6	14.8	13.7	10.9	7.4	4.0	2.6		
		MCDB	3.0	3.4	3.9	7.0	13.1	15.8	19.5	17.1	13.4	9.2	5.4	4.1		
	10%	WB	0.5	1.0	1.1	3.3	7.3	10.4	13.7	12.8	10.0	6.4	2.9	1.6		
		MCDB	1.7	2.4	2.8	5.5	11.0	14.0	17.8	15.8	12.2	8.1	4.3	3.0		
(35) Mean Daily Temperature Range	MDBR		3.7	4.0	4.5	5.5	6.1	6.0	6.8	6.3	5.2	3.8	3.4	3.7		
	5% DB	MCDBR	4.2	4.5	4.1	7.2	10.0	10.1	10.8	9.3	7.9	4.9	4.4	4.7		
		MCWBR	3.5	3.5	3.0	4.2	5.2	5.2	5.3	4.9	4.4	3.5	3.3	3.7		
	5% WB	MCDBR	4.1	4.5	4.0	6.6	9.5	9.0	9.9	8.6	6.9	4.5	4.2	4.5		
		MCWBR	3.6	3.7	3.2	4.1	5.1	4.9	5.3	4.8	4.2	3.5	3.4	3.8		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.268	0.316	0.333	0.345	0.343	0.347	0.335	0.316	0.275	N/A	N/A		
	taud		N/A	2.051	2.113	2.227	2.381	2.466	2.488	2.506	2.513	2.197	N/A	N/A		
	Ebn at Noon		N/A	442	669	795	838	853	835	803	714	439	0	0		
	Edh at Noon		N/A	43	81	99	96	91	87	76	57	36	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.01	0.26	1.35	3.18	4.59	4.80	4.35	3.09	1.63	0.50	0.00	0.00		
	RadStd		0.00	0.06	0.19	0.31	0.30	0.62	0.45	0.26	0.19	0.08	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page